

Name: _____

Roll Number: _____

Quiz-2

Max. Time: 12 mins

Max. Marks: 10

Note: Solve all parts. Limit your written responses to the provided space.

Q.1. [5] Choose by putting a check mark on the most appropriate option. Note: No cutting/overwriting allowed.

i. For two player games, 'MIN' can be the root node.

☒ (A) True (B) False

ii. ' α ' variable associated with MIN can only:

(A) Strictly Increase (B) Strictly Decrease (C) Monotonically increase ☒ (D) None of the given options

iii. If it's MIN's move and the maximum advantage it currently has is v , then for a MAX node above the ply depth, we would _____:

(A) Not prune, if $\alpha > v$ (B) Prune, if $v > \alpha$ (C) Not prune, if $\alpha \leq v$ ☒ (D) Prune, if $v \leq \alpha$

iv. β -cut off is when for a current MIN node ' n ', $\beta(n) \leq \alpha(i)$, for some MAX ancestor ' i ' of ' n '.

(A) True ☒ (B) False

v. In the worst case, α - β is as inefficient as minimax.

☒ (A) True (B) False

Q.2. [2]

a) Give the number of nodes explored by the α - β search in the best as well as the worst case.

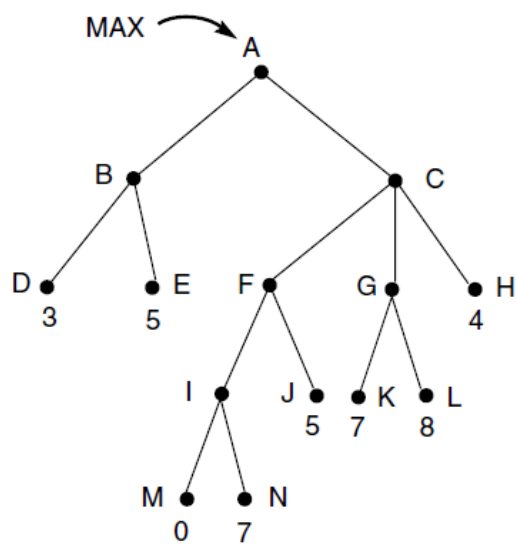
Solution: For b and d denoting the average branching factor and the ply depth, the number of nodes explored is given as follows.

$$\text{Worst Case: } N = \frac{1-b^{(d+1)}}{1-b}$$

$$\text{Best Case: } N = \begin{cases} b^{(d+1)/2} + b^{(d-1)/2} - 1; & \text{Odd ply depth} \\ 2b^{d/2} - 1; & \text{Even ply depth} \end{cases}$$

Q.3. [3]

For the following game tree, perform α - β search from right to left showing the values of the pair as required below.



Solution:

Right to Left

Node	(α,β)
A	$(-i,+i),(4,+i)$
C	$(-i,+i),(-i,4)$
G	$(-i,4),(8,4)$
F	$(-i,4),(5,4)$
B	$(4,+i),(4,5),(4,3)$